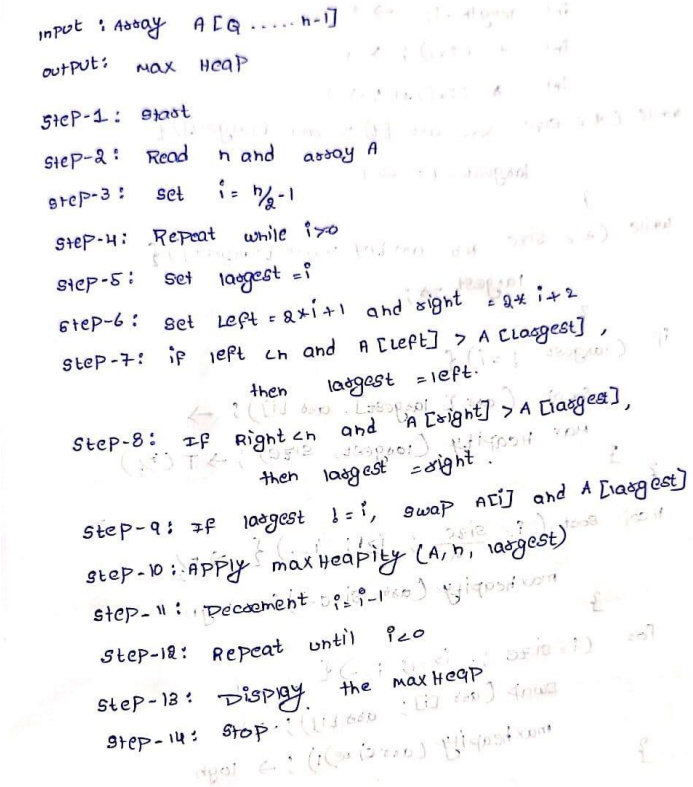
# Practical - 3

**Aim:** Implementation of max-heap sort algorithm

## Max-heap sort

### Algorithm:




### Program:

```
#include <iostream>
using namespace std;
class MaxHeap
{
public:
   int arr[100], n;
   void arrayInput()
   {
      cout << "Enter number of elements: ";
      cin >> n;
      cout << "Enter the elements: ";
      for (int i = 0; i < n; i++)
      {
         cin >> arr[i];
      }
   }
```

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```
    void printArray()
    {
        for (int i = 0; i < n; i++)
        {
            cout << arr[i] << " ";
        }
        cout << endl;
    }

    void maxheapify(int arr[], int n, int i)
    {
        int largest = i;
        int left = 2 * i + 1;
        int right = 2 * i + 2;

        if (left < n && arr[left] > arr[largest])
        {
            largest = left;
        }

        if (right < n && arr[right] > arr[largest])
        {
            largest = right;
        }

        if (largest != i)
        {
            swap(arr[i], arr[largest]);
            maxheapify(arr, n, largest);
        }
    }

    void heapsort()
    {
        for (int i = n / 2 - 1; i >= 0; i--)
        {
            maxheapify(arr, n, i);
        }
        for (int i = n - 1; i > 0; i--)
        {
            swap(arr[0], arr[i]);
            maxheapify(arr, i, 0);
        }
    }
};

int main()
{
    MaxHeap mh;
```

```
    mh.arrayInput();

    cout << "\nBefore Sorting: ";
    mh.printArray();

    mh.heapsort();

    cout << "After Sorting: ";
    mh.printArray();

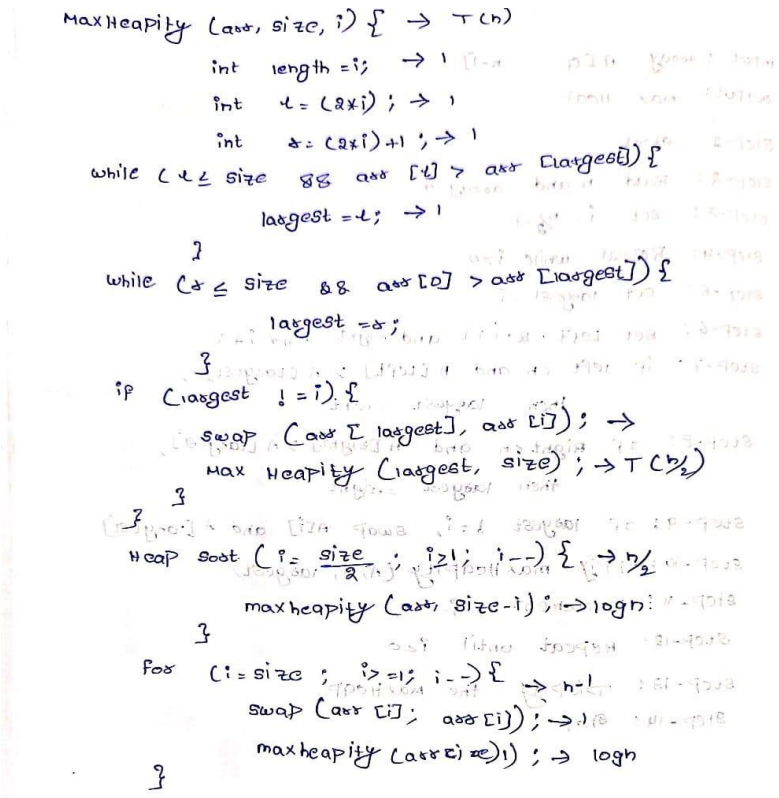
    return 0;
}
```

**Output:**

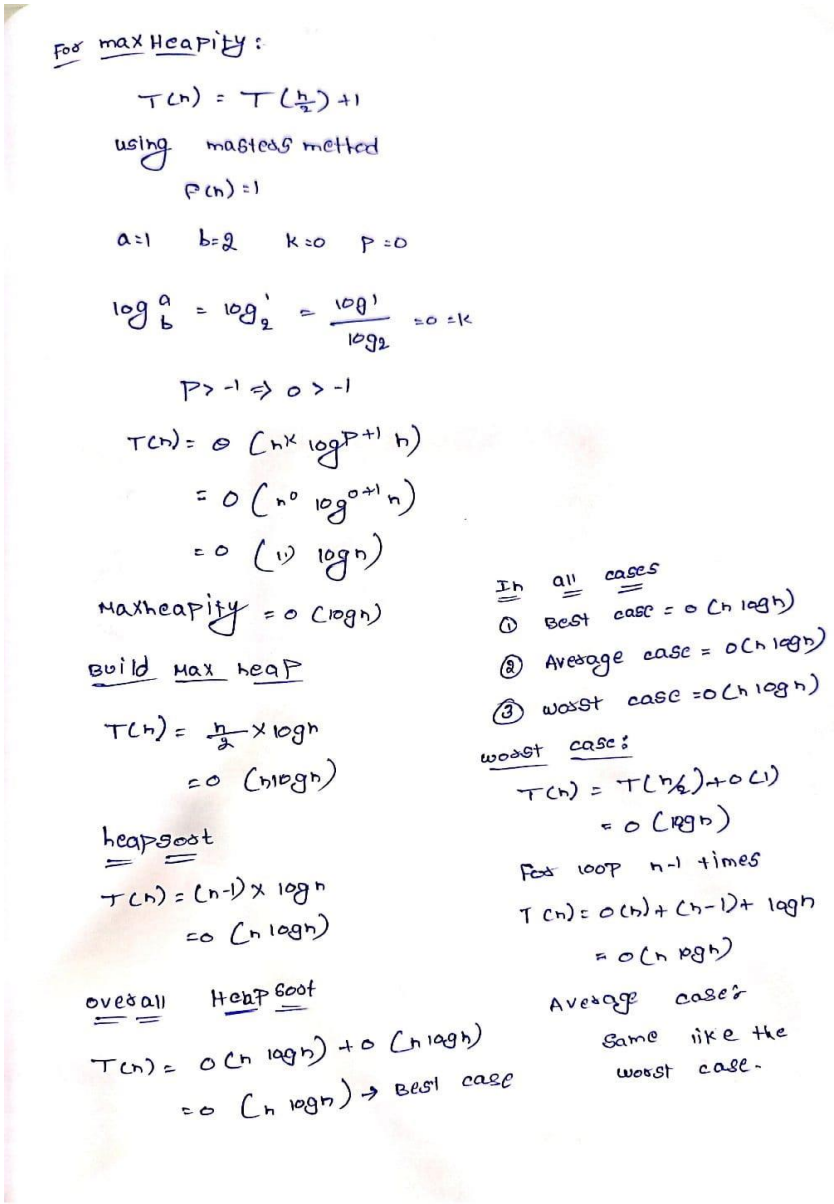
```
PS C:\MinGW\bin> & "C:\Users\chinn\Downloads\dsa in c++\heapify.exe"
Enter the elements: 12
5
3
26
38
9

Before Sorting: 12 5 3 26 38 9
After Sorting: 3 5 9 12 26 38
```

**Time Complexity:**




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**Conclusion:**

The Max-Heap Sort algorithm was successfully implemented using the Max-Heapify method. The largest element is repeatedly placed at the end of the array, resulting in a sorted array in ascending order. The time complexity of Heap Sort is O(n log n).

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